Chengyu Huang

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Education

Xidian University **Xi'an, China**
Undergraduate in Artificial Intelligence *Sep. 2024 – Jun. 2028 (expected)*

- **GPA: 3.9/4.0** **CET-6:** 615
- **Rank: 1/283** (Computer (Intelligence) Category, 2024–2025) **1/153** (Artificial Intelligence)
- **Core Courses:** Advanced Mathematics A(I) (99), Programming Fundamentals B (99), Linear Algebra A (99), Probability Theory (98), Physics (I) (98), Data Structures (97), Discrete Mathematics (96), Advanced Mathematics A(II) (95), Introduction to Artificial Intelligence (94), Mathematical Foundations of Artificial Intelligence (92).

Research Experience

Tsinghua University
Research Intern *Aug. 2026 – Present*

- Conducting research on 4D reconstruction.

Federated Learning **Xidian University**
Undergraduate Researcher *Sep. 2025 – Jul. 2026*

- **Advisor: Xu Zhang**
- Studied the foundations of artificial intelligence and machine learning, focusing on federated learning, personalized learning, and robust learning under label noise.

Publication

- **Personalized Federated Learning with Noisy Labels via Total Variation Regularization**
  Chengyu Huang, Xu Zhang
  We propose pFedTV and cFedTV, which combine total variation regularization, noise transition modeling, and adaptive client clustering for robust personalized federated learning with heterogeneous data and noisy labels.
  Manuscript under review, 2026.

Selected Projects

PaperUnfold *May 2026 – Jun. 2026*

- Developed the initial concept, workflow requirements, and evaluation criteria for PaperUnfold, a Codex skill for deep analysis of AI papers.
- Designed its outputs, including Chinese reading reports, condensed PDFs, and editable diagrams of paper structure and technical lineage, while separating source evidence from analysis and external verification.
  [code]

The Listening Storyteller: An Interactive Storytelling System Powered by LLMs and Expressive Speech Synthesis *Mar. 2026 – Jul. 2026*

- Proposed the project concept and led its design and development; built the frontend with HTML, CSS, and JavaScript and implemented the core backend functions in Python using large model APIs.
- Developed an integrated voice interaction pipeline supporting speech recognition, semantic and emotional understanding, story matching, streaming narration, voice activity detection, and spoken interruption during playback.

[code]

AI Image Style Studio *Jun. 2026 – Jul. 2026*

- Built a Streamlit web application with seven OpenCV artistic filters, image upload, original-result comparison, and PNG download.

[code]

Remote Sensing Image Change Detection Using the Segment Anything Model (SAM) *Apr. 2025 – Oct. 2025*

- Participated in a provincial undergraduate innovation project, assisting with reproduction of the SAM-SCD method and data preprocessing for remote sensing image change detection.

Awards

- **National Scholarship** **2025**
- **Outstanding Student, Xidian University** **2025**
- **Provincial First Prize, Contemporary Undergraduate Mathematical Contest in Modeling** **2025**